

AtliQ Hardwares - Sales & Profitability Analysis Report

1. Executive Summary

This report presents an end-to-end data analytics workflow for AtliQ Hardwares, a computer hardware and peripherals manufacturer. The initiative transformed raw transactional records stored in a MySQL database into a dynamic, executive-level Power BI report. The analysis identifies key revenue drivers, regional margin variations, and critical data quality issues to support strategic decision-making.

2. Business Problem & Objectives

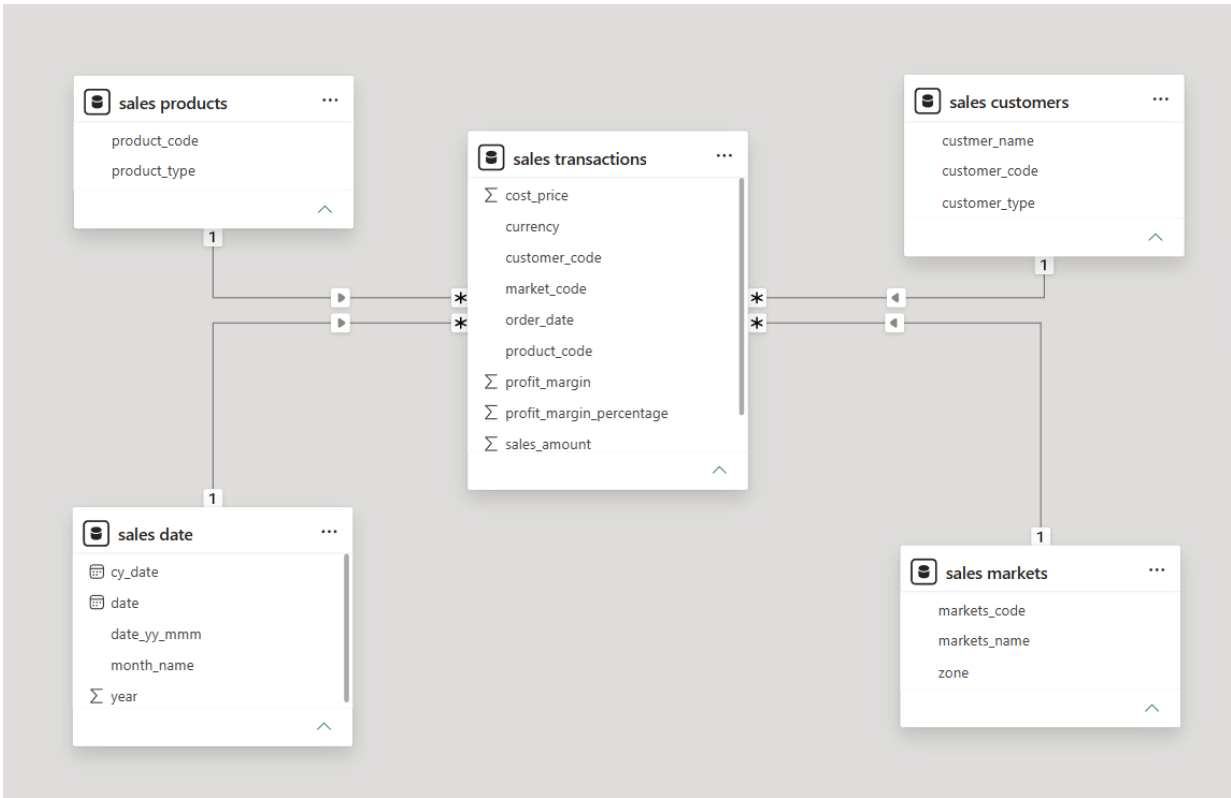
- **Centralized Visibility:** Lack of a unified view for tracking regional revenue and sales volume.
- **Profitability Monitoring:** Difficulty in evaluating profit margin contributions across different customer segments and products.
- **Target Tracking:** Need for dynamic scenario analysis to evaluate regional performance against custom profit margin targets.

3. Data Architecture & Modeling

The project uses a **Star Schema** built inside Power BI, centered around the transactions fact table linked to four core dimension tables:

Table Name	Table Type	Description & Key Fields
sales transactions	Fact Table	Contains order-level transactions, quantity, currency, and profit margins.
sales customers	Dimension Table	Customer names, market mapping, and customer codes.
sales products	Dimension Table	Product IDs, product categories, and codes.
sales markets	Dimension Table	Regional market codes, market names, and geographical zones.

Table Name	Table Type	Description & Key Fields
sales date	Dimension Table	Order dates, month names, quarters, and years.

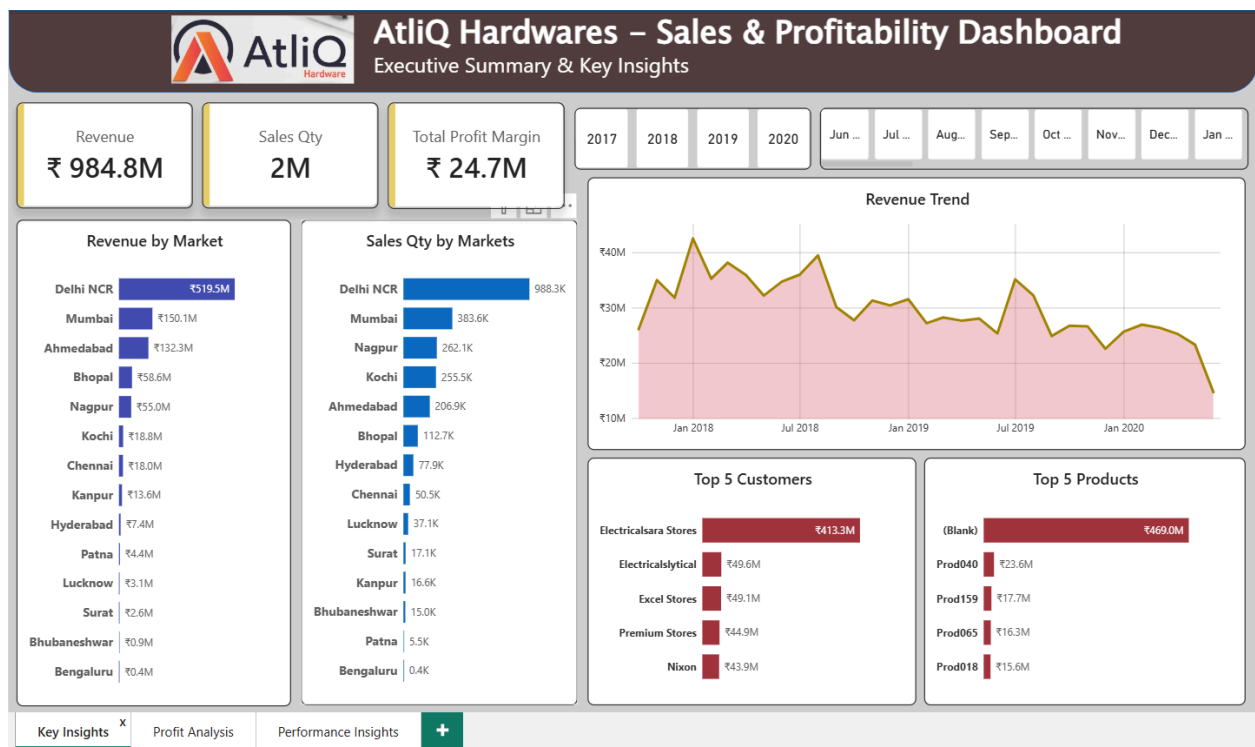


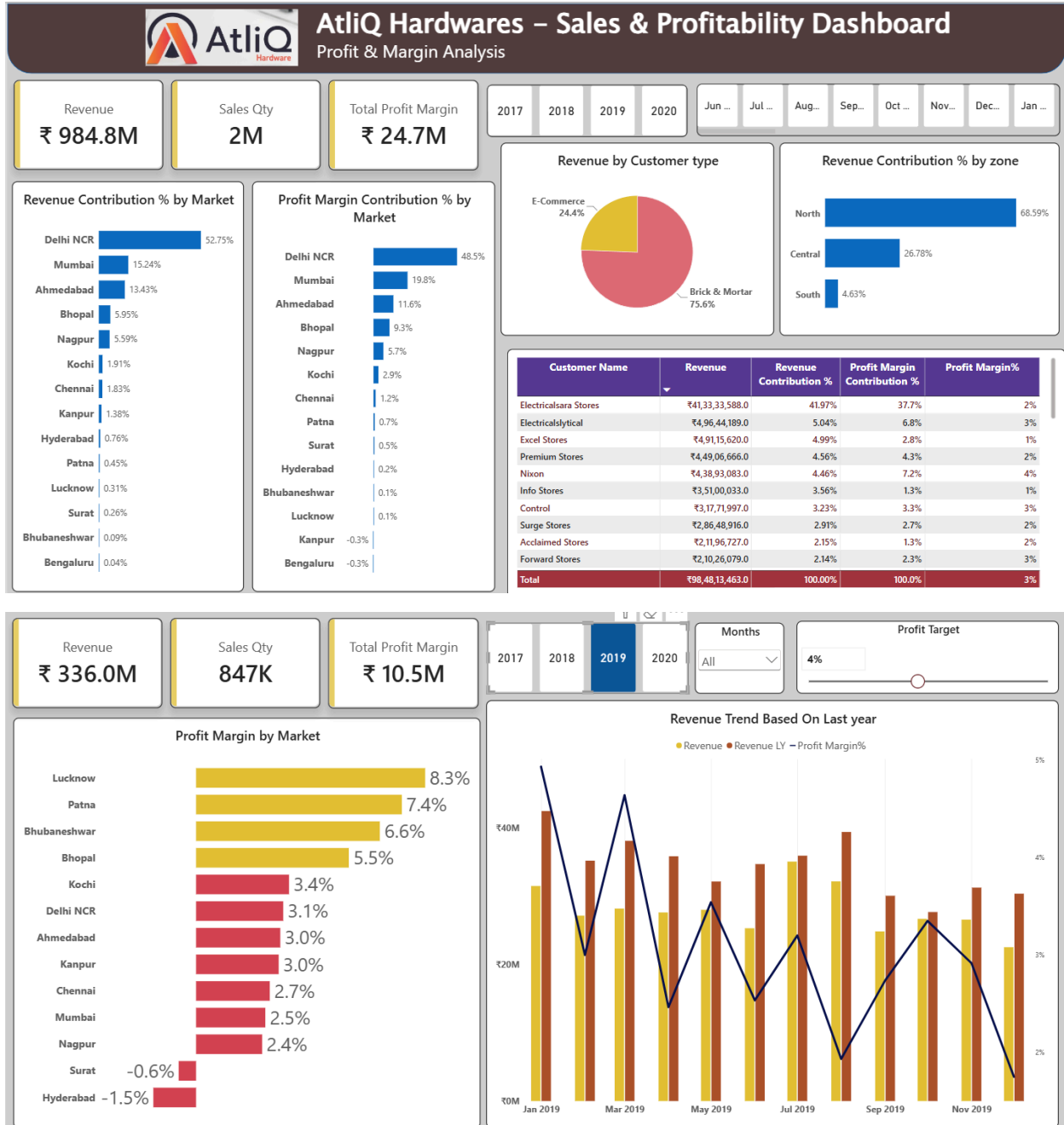
4. Interactive Dashboard Features

The interactive Power BI dashboard is structured to deliver immediate visual insights for executive decision-making across three primary perspectives:

Dashboard Visual / Component	Functional Purpose	Key Interactivity & Insights
Executive KPI Cards	High-level performance summary	Displays real-time totals for Revenue , Total Profit Margin , Sales Quantity , and overall Profit Margin % .
Regional Revenue & Margin	Geographic performance	Bar and column charts showing

Dashboard Visual / Component	Functional Purpose	Key Interactivity & Insights
Breakdown	mapping	top revenue-generating markets (e.g., Delhi NCR, Mumbai) vs. actual regional profit margins.
Profit Target What-If Slicer	Dynamic scenario modeling	A custom parameter slider (GENERATESERIES -5% to 15%) allowing leadership to adjust profit expectations and instantly highlight underperforming markets via Target Diff.
Top 5 Products & Customers Breakdown	Product & client ranking	Visual ranking of top-performing SKUs and clients by revenue contribution percentage.





Revenue

₹ 336.0M

Sales Qty

847K

Total Profit Margin

₹ 10.5M

2017

2018

2019

2020

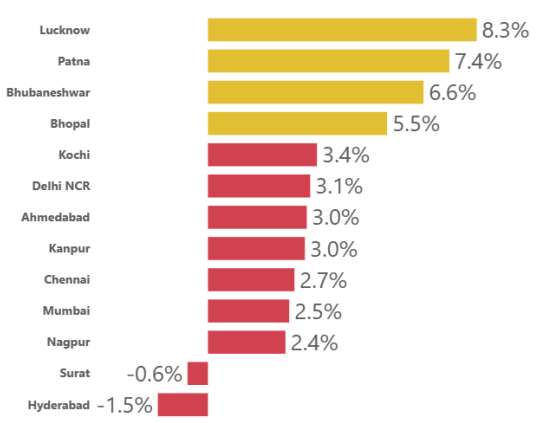
Months

All

Profit Target

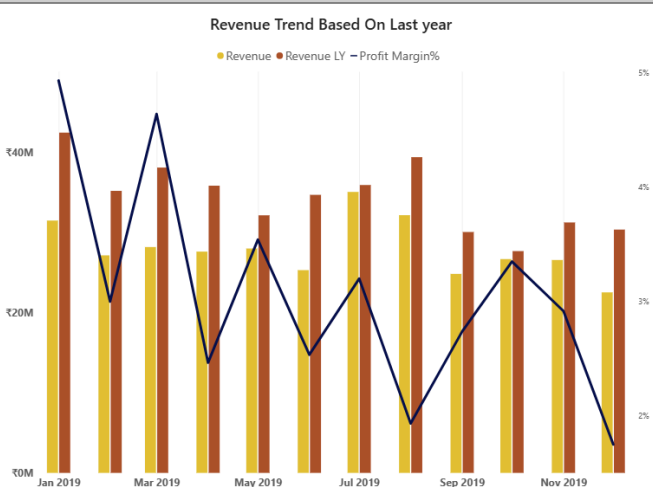
4%

Profit Margin by Market



Market	Profit Margin %
Lucknow	8.3%
Patna	7.4%
Bhubaneswar	6.6%
Bhopal	5.5%
Kochi	3.4%
Delhi NCR	3.1%
Ahmedabad	3.0%
Kanpur	3.0%
Chennai	2.7%
Mumbai	2.5%
Nagpur	2.4%
Surat	-0.6%
Hyderabad	-1.5%

Revenue Trend Based On Last year



5. Technical Implementation & Logic

A. Exploratory Data Analysis (SQL)

1. Filter 2020 transactional records

```

SELECT transactions.*, date.*
FROM transactions
INNER JOIN date ON transactions.order_date = date.date
WHERE date.year = 2020;

```

2. Audit total 2020 revenue considering currency strings

```
SELECT SUM(transactions.sales_amount)
FROM transactions
INNER JOIN date ON transactions.order_date = date.date
WHERE date.year = 2020 AND (transactions.currency = "INR\r" OR transactions.currency = "USD\r");
```

B. Key DAX Calculations

- **Revenue:** Revenue = SUM('sales transactions'[sales_amount])
- **Total Profit Margin:** Total Profit Margin = SUM('sales transactions'[profit_margin])
- **Profit Margin %:** Profit Margin% = DIVIDE([Total Profit Margin], [Revenue], 0)
- **Revenue Contribution %:** Revenue Contribution % = DIVIDE([Revenue], CALCULATE([Revenue], ALL('sales products'), ALL('sales customers'), ALL('sales markets')))
- **Target Difference:** Target Diff = [Profit Margin%] - 'Profit Target'[Profit Target Value]

6. Key Business Insights & Findings

- **Top Revenue Drivers:** Key metro markets like Delhi NCR and Mumbai contribute the majority of total sales revenue.
- **Margin Discrepancy:** High-volume sales in specific regions do not directly correlate with high profitability due to lower unit margins.
- **What-If Target Sensitivity:** Raising the dynamic profit target above 3% highlights multiple regional markets operating below threshold profitability.

7. Critical Data Anomaly Identified

Issue Category	Description	Impact & Recommendation
Data Quality / Source Error	In the Top 5 Products analysis, the highest-ranking entry displays as Blank / Unlabeled .	This stems from a source data error at AtliQ Hardwares where transactional records lack corresponding product IDs in the master product table. It is recommended that AtliQ's database team conduct a database audit to map missing SKU IDs.